IOC Lookup Tool

A small Python tool I built to check IPs, file hashes, domains, URLs, and email headers against AbuseIPDB and VirusTotal — instead of doing it manually one at a time.

Why I built this

I was working through a home SOC lab (Wazuh + Atomic Red Team + OpenVAS) and kept manually pasting IOCs into different websites to check them. It got old fast, especially when checking more than a couple at once. I looked into how SOAR platforms automate this and decided to build a scaled-down version myself — partly to save time, mostly to actually understand what's happening instead of just clicking buttons in a tool.

It started as a single-IP checker. Once that worked, I added hashes, domains, URLs, and email header analysis one at a time, testing each before moving to the next.

What it does

- Checks IPs against AbuseIPDB (abuse score, country, ISP, report categories)
- Checks hashes, domains, and URLs against VirusTotal (detection ratio, tags, WHOIS for domains)
- Parses raw email headers — pulls sender IP, From domain, SPF/DKIM/DMARC results, and flags Reply-To mismatches to catch phishing
- Auto-detects what type of IOC you gave it, so you don't have to specify
- Two ways to run it: a terminal version for quick checks, and an Excel version that saves results and color-codes them (red/orange/green) so you can scan a batch of results at a glance

How it works (short version)

You give it an IOC → it figures out what type it is → sends it to the right API → reads back the JSON response → applies some simple threshold logic (e.g. 5+ AV engines flagging something = malicious) → prints or writes the result.

Tech used

Python, requests, openpyxl, AbuseIPDB API, VirusTotal API v3

Screenshots

Results from real test runs, both the PowerShell and Excel versions, across all supported IOC types.

Why two versions?

I initially built the PowerShell version, but results only printed to the terminal — nothing was saved, and reviewing more than a few results at once was hard to track. That's when I had the idea to move the same logic into Excel, so results would persist and be easier to scan visually with color coding. Both versions are shown here to demonstrate that progression.

Batch check — mixed IPs, hashes, domains, and URLs in one run (Powershell):

```
[1/8] Checking: 185.220.101.45
==================================================
IOC                  : 185.220.101.45
Type                 : IP
Verdict              : MALICIOUS
Score                : 99%
Country              : DE
ISP                  : Network for Tor-Exit traffic.
Report Categories    : Hacking, SQL Injection, Brute-Force, Bad Web Bot, Web App Attack, SSH
Last Reported        : 2026-08-02
==================================================
[2/8] Checking: 45.155.205.233
==================================================
IOC                  : 45.155.205.233
Type                 : IP
Verdict              : CLEAN
Score                : 6%
Country              : RU
ISP                  : Cloud Technologies LLC trading as Cloud.ru
Report Categories    : Brute-Force, SQL Injection, SSH, Hacking
Last Reported        : 2026-07-31
==================================================
[3/8] Checking: 44d88612fea8a8f36de82e1278abb02f
C:\Users\jasmin\Documents\Python\IP_lookup_v5.py:60: DeprecationWarning: datetime.datetime.utcfromtimestamp() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.fromtimestamp(timestamp, datetime.UTC).
  return datetime.utcfromtimestamp(timestamp).strftime("%Y-%m-%d")
==================================================
IOC                  : 44d88612fea8a8f36de82e1278abb02f
Type                 : Hash
Verdict              : MALICIOUS
Detections           : 53/75 engines
First Seen           : 2006-05-22
Last Seen            : 2026-08-02
Tags                 : via-tor, direct-cpu-clock-access, idle, long-sleeps, powershell, detect-debug-environment, attachment, known-distributor
==================================================
[4/8] Checking: da39a3ee5e6b4b0d3255bfef95601890afd80709
==================================================
IOC                  : da39a3ee5e6b4b0d3255bfef95601890afd80709
Type                 : Hash
Verdict              : CLEAN
Detections           : 0/75 engines
First Seen           : 2006-09-18
Last Seen            : 2026-08-02
Tags                 : software-collection, known-distributor, nsrl, runtime-modules, zero-filled, direct-cpu-clock-access, via-tor, trusted, attachment, legit
==================================================
[5/8] Checking: cim-kolea.com
==================================================
IOC                  : cim-kolea.com
Type                 : Domain
Verdict              : SUSPICIOUS
Detections           : 3/91 engines
Last Seen            : 2026-08-02
Tags                 : None
WHOIS Registrar      : NAMECHEAP INC
Domain Created       : 2019-10-22T17:10:16.00Z
==================================================
[6/8] Checking: google.com
==================================================
IOC                  : google.com
Type                 : Domain
Verdict              : CLEAN
Detections           : 0/91 engines
Last Seen            : 2026-08-02
Tags                 : None
WHOIS Registrar      : MarkMonitor Inc.
Domain Created       : 1997-09-15T04:00:00Z
==================================================
[7/8] Checking: http://testsafebrowsing.appspot.com/s/malware.html
==================================================
IOC                  : http://testsafebrowsing.appspot.com/s/malware.html
Type                 : URL
Verdict              : MALICIOUS
Detections           : 11/92 engines
First Seen           : 2016-10-02
Last Seen            : 2026-08-02
Tags                 : contains-zip
==================================================
[8/8] Checking: https://www.wikipedia.org
==================================================
IOC                  : https://www.wikipedia.org
Type                 : URL
Verdict              : CLEAN
Detections           : 0/92 engines
First Seen           : 2012-06-15
Last Seen            : 2026-08-01
Tags                 : None
==================================================

SUMMARY
--------------------------------------------------
185.220.101.45                                     -> MALICIOUS
45.155.205.233                                     -> CLEAN
44d88612fea8a8f36de82e1278abb02f                   -> MALICIOUS
da39a3ee5e6b4b0d3255bfef95601890afd80709           -> CLEAN
cim-kolea.com                                      -> SUSPICIOUS
google.com                                         -> CLEAN
http://testsafebrowsing.appspot.com/s/malware.html -> MALICIOUS
https://www.wikipedia.org                          -> CLEAN
```

IP Sheet (Excel)

IOC	Verdict	Score	Country	ISP	Report Categories	Last Reported
8.8.8.8	CLEAN	0%	US	Google LLC	Hacking, SQL Injection, Brute-Force, Exploited Host, SSH	2026-08-02
1.1.1.1	CLEAN	0%	AU	APNIC and Cloudflare DNS Resolver project	Brute-Force, SSH, Hacking, SQL Injection	2026-08-02
185.220.101.1	MALICIOUS	100%	DE	Artikel10 e.V.	Web Spam, Hacking, SQL Injection, Brute-Force, Bad Web Bot, Exploited Host, Web App Attack, SSH	2026-08-02
185.220.101.45	MALICIOUS	99%	DE	Network for Tor-Exit traffic.	Hacking, SQL Injection, Brute-Force, Bad Web Bot, Web App Attack, SSH	2026-08-02
45.155.205.233	CLEAN	6%	RU	Cloud Technologies LLC trading as Cloud.ru	Brute-Force, SQL Injection, SSH, Hacking	2026-07-31
115.49.66.100	CLEAN	0%	CN	China Unicom Henan province network	None	Never
46.227.184.200	CLEAN	15%	KZ	Kar-Tel LLC	Hacking, SQL Injection	2026-07-29
218.56.204.114	SUSPICIOUS	33%	CN	China Unicom Shandong province network	Brute-Force, Hacking, IoT Targeted	2026-08-02

Hash Sheet (Excel)

IOC	Verdict	Detections	First Seen	Last Seen	Tags
44d88612fea8a8f36de82e1278abb02f	MALICIOUS	65/75 engines	2006-05-22	2026-08-02	idle, detect-debug-environment, attachment, direct-cpu-clock-access, long-sleeps, powershell, via-tor, known-distributor
3395856ce81f2b7382dee72602f798b642f14140	MALICIOUS	65/75 engines	2006-05-22	2026-08-02	idle, detect-debug-environment, attachment, direct-cpu-clock-access, long-sleeps, powershell, via-tor, known-distributor
131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbdfd8267	MALICIOUS	62/75 engines	2006-05-23	2026-07-31	idle, detect-debug-environment, powershell, known-distributor, attachment, long-sleeps, via-tor
275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0	ERROR	HTTP 400			
d41d8cd98f00b204e9800998ecf8427e	CLEAN	0/75 engines	2006-09-18	2026-08-02	direct-cpu-clock-access, runtime-modules, attachment, software-collection, trusted, known-distributor, via-tor, zero-filled, legit, nsrl
da39a3ee5e6b4b0d3255bfef95601890afd80709	CLEAN	0/75 engines	2006-09-18	2026-08-02	direct-cpu-clock-access, runtime-modules, attachment, software-collection, trusted, known-distributor, via-tor, zero-filled, legit, nsrl
e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855	CLEAN	0/75 engines	2006-09-18	2026-08-02	direct-cpu-clock-access, runtime-modules, attachment, software-collection, trusted, known-distributor, via-tor, zero-filled, legit, nsrl
5d41402abc4b2a76b9719d911017c592	CLEAN	0/74 engines	2007-02-11	2026-07-22	text, trusted, nsrl, known-distributor, attachment, long-sleeps, idle
2aae6c35c94fcfb415dbe95f408b9ce91ee846ed	CLEAN	0/74 engines	2009-03-15	2026-07-15	long-sleeps, trusted, legit, known-distributor, idle, text
2cf24dba5fb0a30e26e83b2ac5b9e29e1b161e5c1fa7425e73043362938b9824	CLEAN	0/74 engines	2007-02-11	2026-07-22	text, trusted, nsrl, known-distributor, attachment, long-sleeps, idle
7bf2b57f2a205768755c07f238fb32cc	MALICIOUS	64/74 engines	2017-05-12	2026-07-27	peexe, checks-user-input, runtime-modules, via-tor, direct-cpu-clock-access
2ef7bde608ce5404e97d5f042f95f89f0c34e6f9	NOT FOUND				
7c211433f02071597741e6ff5a8ea34789abbf43729ba69007ed81c17c04d47	ERROR	HTTP 400			
b026324c6904b2a9cb4b88d6d61c81d1	CLEAN	0/75 engines	2009-02-26	2026-08-02	idle, trusted, text, known-distributor, via-tor, long-sleeps, legit, nsrl
0cc175b9c0f1b6a831c399e269772661	CLEAN	0/74 engines	2008-05-21	2026-07-21	text, legit, nsrl, attachment, known-distributor, via-tor

Domain Sheet (Excel)

IOC	Verdict	Detections	Last Seen	Tags	WHOIS Registrar	Domain Created
google.com	CLEAN	0/91 engines	2026-08-0	None	MarkMonitor Inc.	1997-09-15
microsoft.com	CLEAN	0/91 engines	2026-08-0	None	Unknown	Unknown
cloudflare.com	CLEAN	0/91 engines	2026-08-0	None	Cloudflare, Inc.	2009-02-17
github.com	CLEAN	0/91 engines	2026-08-0	None	MarkMonitor Inc.	2007-10-09
wikipedia.org	CLEAN	0/91 engines	2026-08-0	None	Unknown	Unknown
cim-kolea.com	SUSPICIOUS	3/91 engines	2026-08-0	None	NAMECHEAP INC	2019-10-22

URL Sheet (Excel)

IOC	Verdict	Detections	First Seen	Last Seen	Tags
http://google.com	CLEAN	0/92 engines	2010-06-14	2026-08-02	external-resources, multiple-redirects
http://testsafebrowsing.appspot.com/s/malware.html	MALICIOUS	11/92 engines	2016-10-02	2026-08-02	contains-zip
https://github.com	CLEAN	0/92 engines	2011-04-25	2026-08-02	external-resources
https://www.wikipedia.org	CLEAN	0/92 engines	2012-06-15	2026-08-01	None
http://example.com	CLEAN	0/92 engines	2010-10-16	2026-08-02	None

Email Headers Sheet (Excel)

Header Text	Sender IP	From Domain	Reply-To Domain	SPF	DKIM	DMARC	Reply-To Mismatch	Sender IP Verdict	From Domain Verdict	Overall Verdict
Received: from mail-relay.google.com (209.85.220.41)	209.85.220.41	notifications.google.com	notifications.google.com	PASS	PASS	PASS	NO	CLEAN	CLEAN	LOW RISK
Received: from mail-fraud.suspicious-host.ru (203.0.113.77)	203.0.113.77	micros0ft-security.com	fake-support-team.net	FAIL	FAIL	FAIL	YES	CLEAN	CLEAN	HIGH CONFIDENCE PHISHING
